

EMCH380 – Project Management Project Monitoring & Control

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Slides are developed from instructor's notes and other references.

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Introduction

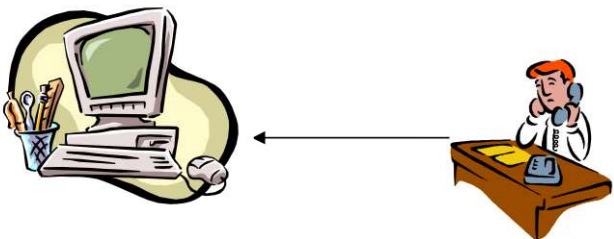
Project Management – Project Monitoring & Control

PROJECT MONITORING & UPDATING

- The initial project plan is just that, an initial plan.
- A project plan soon becomes out-of-date and of minimal use if it is not continually updated and reworked in response to the changing conditions of the project.

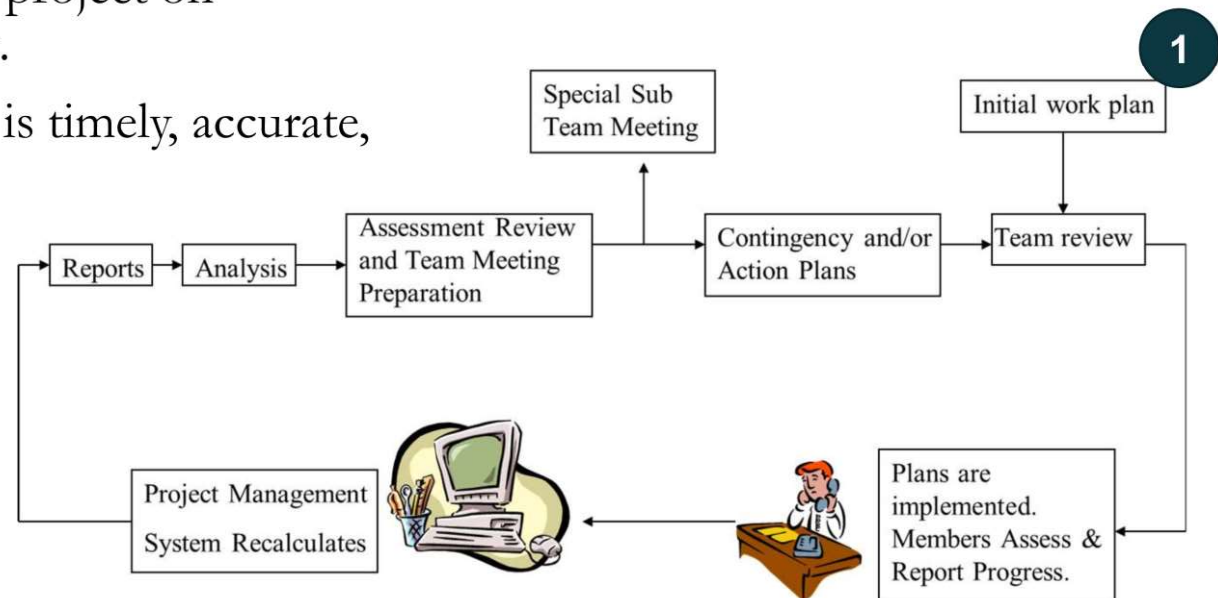
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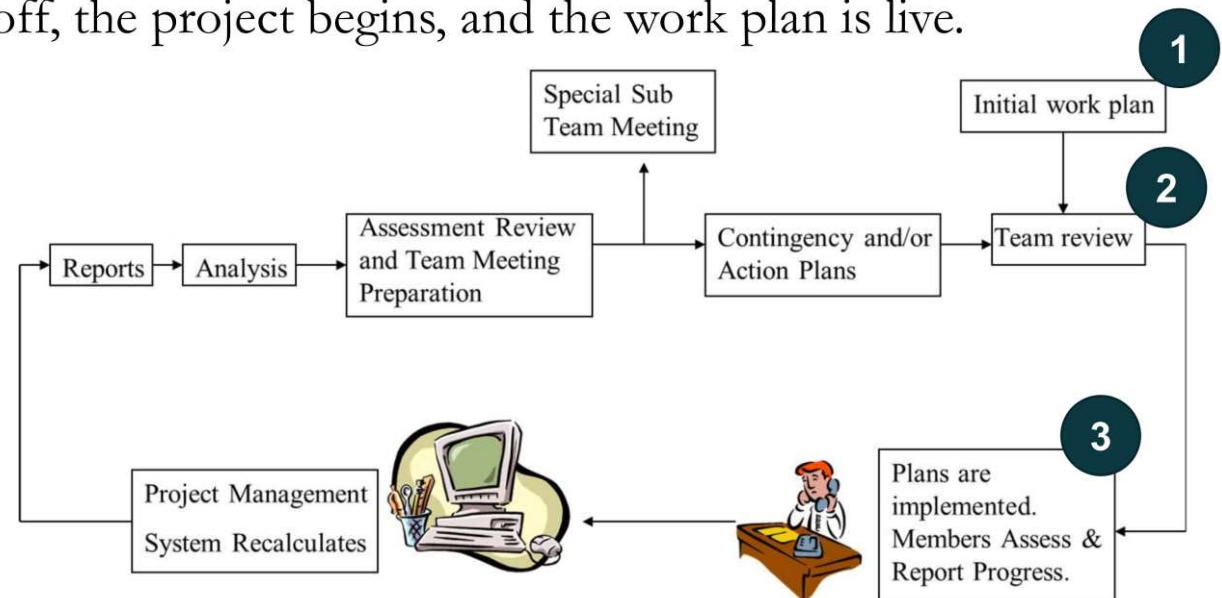
PROJECT MONITORING & UPDATING

- Review the tracking process beginning with initial work plan in the upper left corner.
- Project tracking is a continuous feedback process that helps keep the project on schedule and within budget.
- The key to project tracking is timely, accurate, task progress data.



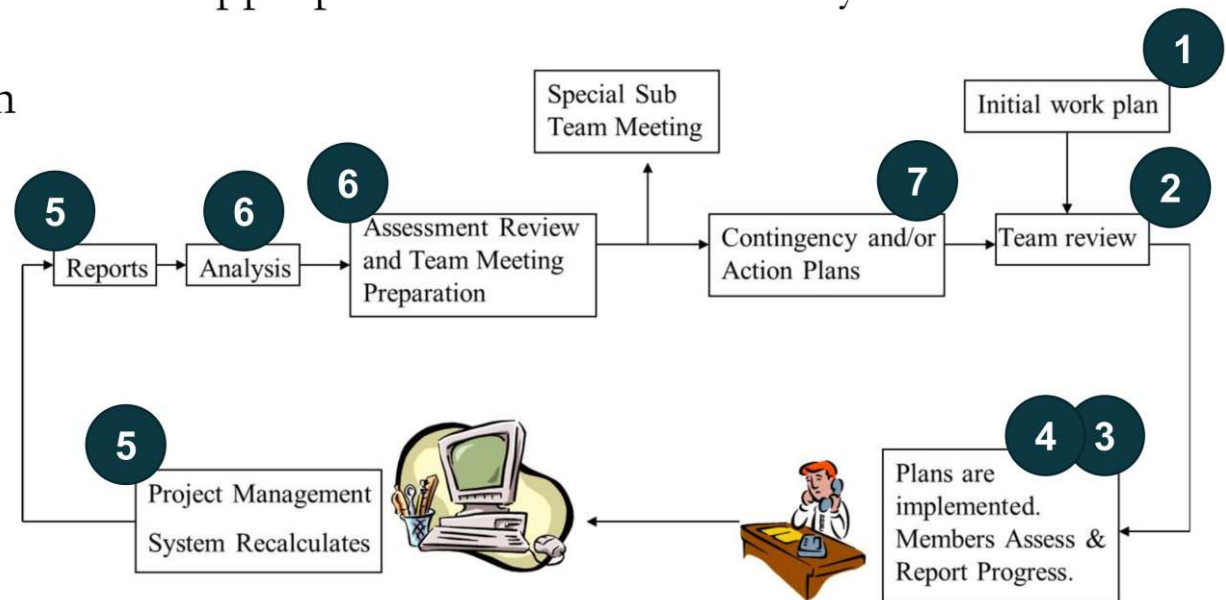
PROJECT MONITORING & UPDATING

- The feedback loop proceeds as follows:
 1. Initial work plan developed (upper right corner of flow chart)
 2. Team review and sign-off
 3. Once initially signed-off, the project begins, and the work plan is live.



PROJECT MONITORING & UPDATING

4. Task progress reported to team leader
5. Team leader/process leader compile all the task progress data and get a project-wide vision of progress.
6. Team leader/process leader and appropriate team members analyze the progress data.
7. Contingency and action plans are developed to resolve problems with tasks.



PROJECT MONITORING & UPDATING

- In this way we constantly monitor the project and stay on top of it. We shouldn't be surprised.
- Problems are well documented. Getting management help will be more efficient and effective.
- We will begin the next section by assuming the plan has just become live, and the team is reporting progress for the first time. (See highlighted area of flow diagram)

I

PROJECT TACKING

- Project Tracking (Working the Schedule)
 - Reporting progress
 - Tracking Reports
 - Analysis / Keeping the project on track
 - Action and Contingency Plan development
- Project Tracking is another way to refer to Project Monitoring and Updating

Supplying the project team with the skills required to keep the plan up-to-date and the project on schedule

PROJECT TRACKING

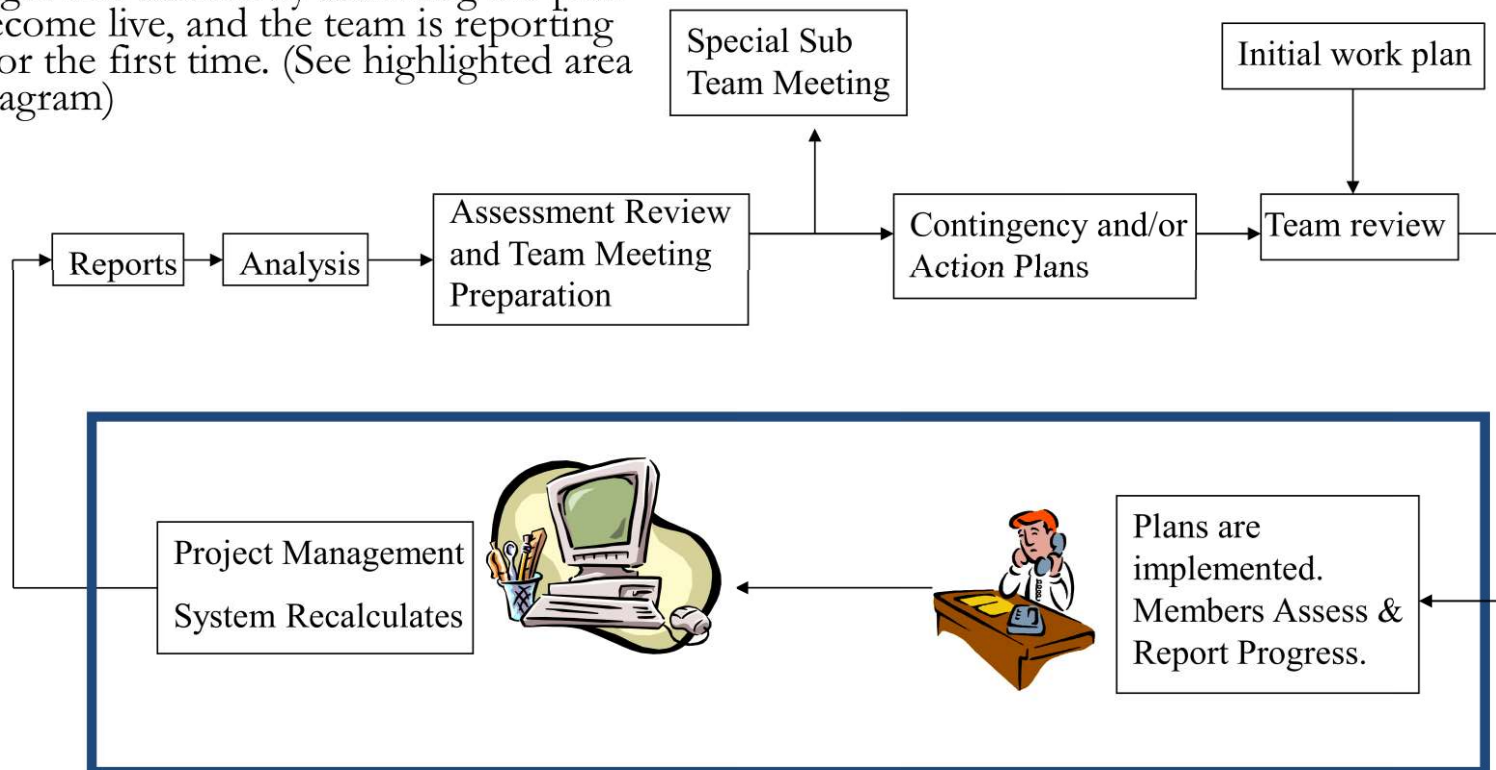
- In this chapter, we will look at Project Tracking
- During the implementation of the plan, feedback is required to determine the amount of work completed and to make adjustments to the plan as required
- Task leaders need to report progress, usually as percent complete, on a weekly basis
- The data from all the task leaders is compiled by the team leader and process leader. They are responsible for analyzing the progress of the entire project. Appropriate task leaders help them analyze particular tasks.
- Action plans are developed to deal with tasks that are behind schedule, experiencing problems, or have an opportunity to improve the work plan.
- All tracking data is maintained and provides an historical record that can be leveraged by subsequent project. This way we don't have to "reinvent the wheel" for every project.
- Accurate data and clear communication are the keys to successful project tracking and, ultimately, to keeping the project on schedule and within budget.

Reporting Progress

Project Management – Project Monitoring & Control

REPORTING PROGRESS

- We will begin this section by assuming the plan has just become live, and the team is reporting progress for the first time. (See highlighted area of flow diagram)



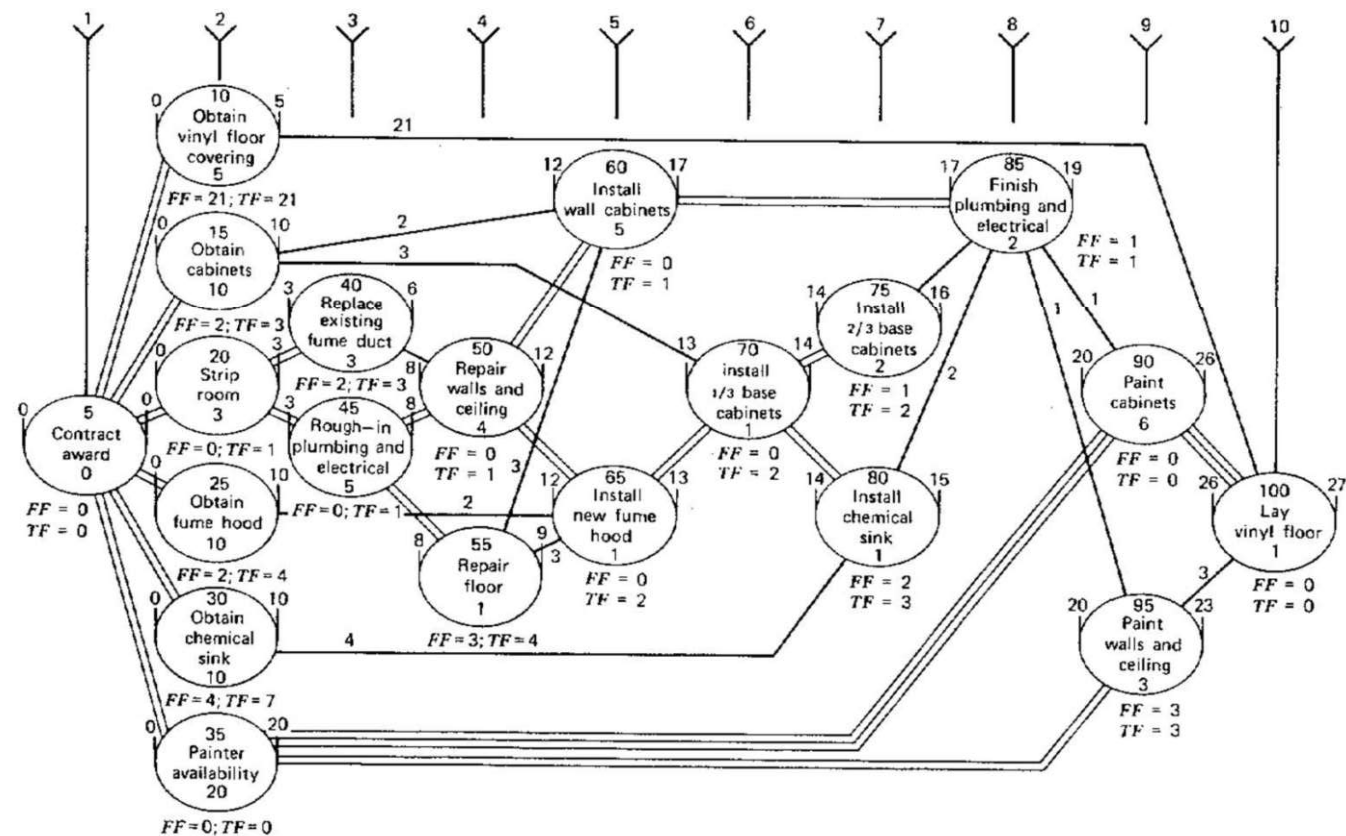
REPORTING PROGRESS

- Task-by-task reporting is the heartbeat of project tracking
- A **task leader** is assigned to each task in the plan. The task leader is responsible for monitoring the progress of the task and reporting task progress to the team leader.
- Task progress is reported every week to the team leader (the data is analogous to Statistical Process Control used in manufacturing). The process data is the project talking back to us. We can see tasks heading out of control before a disaster hits.
- By closely tracking the project (tracking each task), we can react quickly to slippages and problems.
- The team leader is the nerve center for all the data

REPORTING PROGRESS

- Why is it important?
 - React quickly to potential slippages and/or changes in the plan
 - Provides data required to anticipate changes in the plan
- Roles and responsibilities of team members
 - Updates task progress
 - Identifies task slippages
 - Identifies issues
 - Identifies expenses incurred

CHEMICAL LAB EXAMPLE – BASELINE SCHEDULE

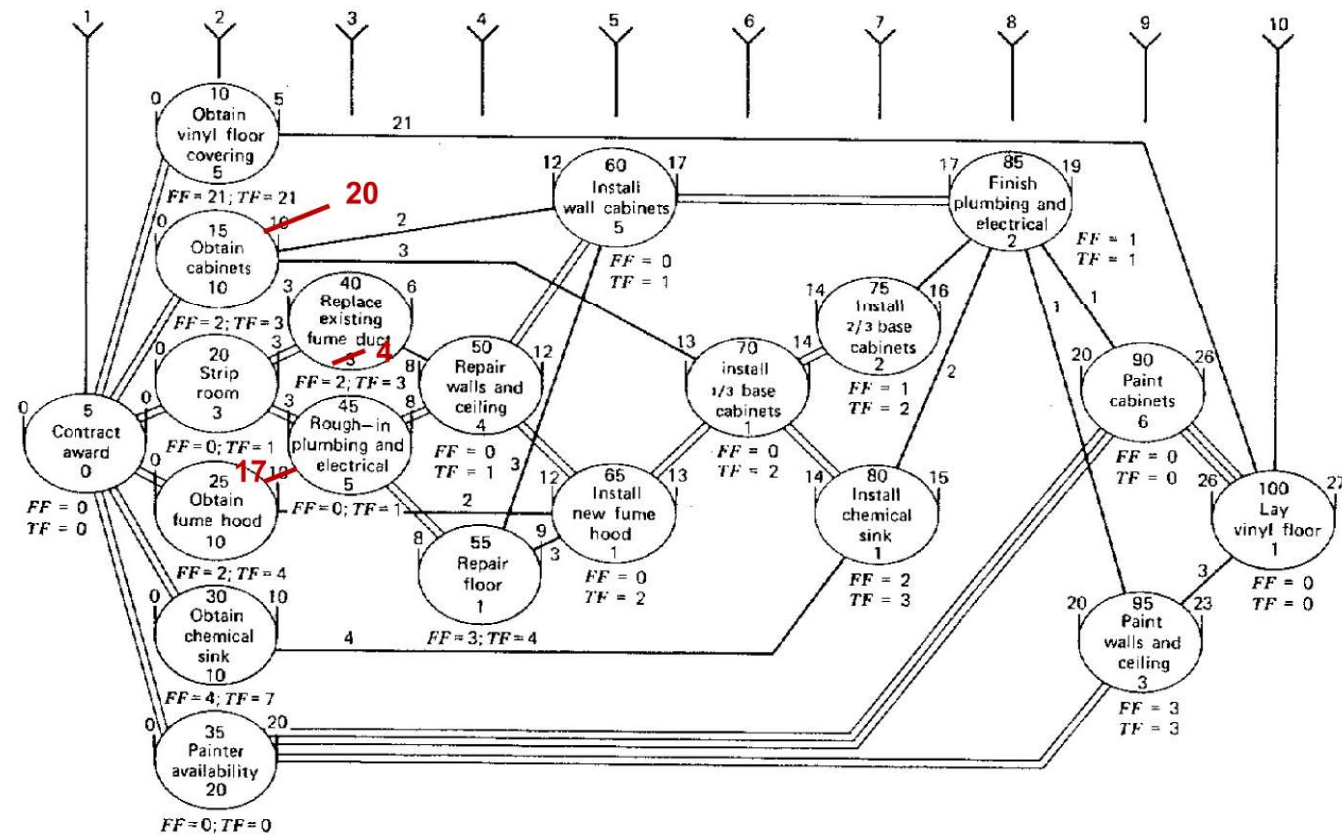


CHEMICAL LAB EXAMPLE

REPORTING PROGRESS

- Work has been underway for 15 days. Reported problems and progress indicate the following:
 - The cabinet supplier reported that it was impossible to deliver all cabinets within 10 days → Procurement problem.
 - Base cabinets are delivered on time.
 - Wall cabinets take longer to build and expected delivery will be delayed and are expected on the 20th day.
 - The fume hood manufacturer needs 17 days to deliver the hood instead of 15 days projected.
 - Replacement of the fume duct took 4 days to complete instead of 3 days, but it is now finished.

CHEMICAL LAB EXAMPLE – BASELINE SCHEDULE

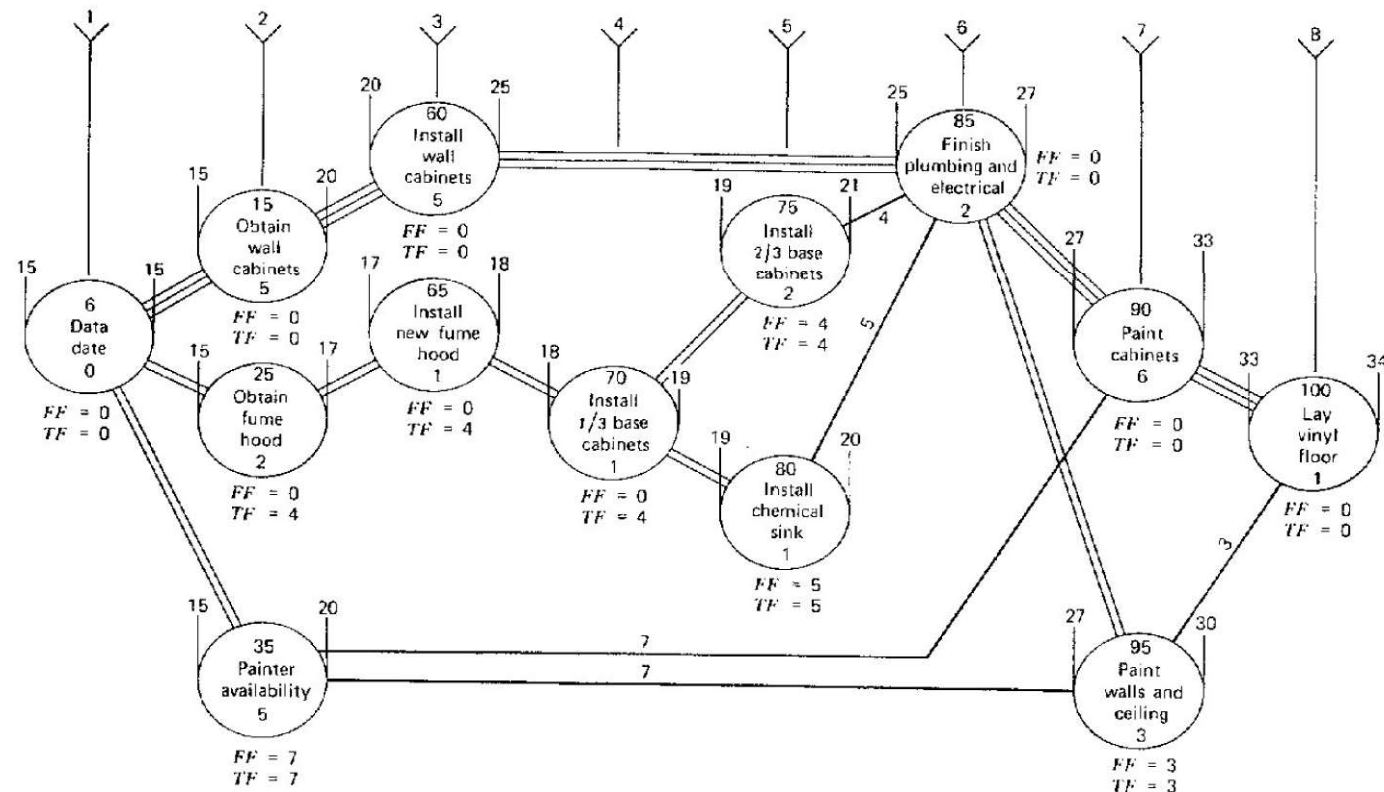


CHEMICAL LAB EXAMPLE

REPORTING PROGRESS

- Impact on Activity durations and planned start and finish dates of activities:
 - Obtain wall cabinets remaining duration = 5
 - Obtain fume hood remaining duration = 2
- Install wall cabinets is delayed because of Act 15 “Obtain cabinets”
- Act 65 “Install new fume hood” is also delayed because Act 25 was delayed. And so on..

CHEMICAL LAB EXAMPLE – UPDATED SCHEDULE



REPORTING PROGRESS

II

- The task leader is responsible for completing the progress data
- Percent complete are estimated by the task leader. Care must be taken when evaluating percent complete
 - As a rule, if the planned time is 50% used up, the task should be approximately 50% complete
- There are exceptions to the rule. The amount of work completed is not necessarily linear with the amount of time used.
- Be careful to use common sense when evaluating percent work complete and amount of time used.

REPORTING PROGRESS

- Recording progress provides the team with the data required to monitor the project and surface issues.
- In addition, it provides a means for collecting historic data to be used in planning the next project.

II

WAYS FOR MEASURING PROGRESS

- Different Methods:
 - Units Completed
 - Incremental Milestone
 - Start / Finish
 - Supervisor Opinion
 - Cost Ratio

WAYS FOR MEASURING PROGRESS

UNITS COMPLETED

- This method is applicable to those tasks involving repeated production of easily measured pieces of work, each requiring approximately the same level of effort
- Example: 10000M of wire is to be pulled and 4000M has been pulled then:

$$\% \text{ Complete} = 4000/10000 = 40\%$$

WAYS FOR MEASURING PROGRESS

INCREMENTAL MILESTONE

- This method is applicable to any activity which includes subtasks that must be handled in sequence.
- Example: Consider the installation of a major vessel in an industrial facility as being the task in question. This task includes the sequential subtasks or operations in the following list:
 - Received and inspected 15%
 - Setting complete 35%
 - Alignment complete 50%
 - Internals installed 75%
 - Testing complete 90%
 - Accepted by owner 100%
- Completion of any operation is considered to be a milestone, which when completed represents a certain percentage of total installation
- The % chosen to represent each milestone is normally based on the number of work-hours estimated to be required to that point in relation to the total.

WAYS FOR MEASURING PROGRESS

START FINISH METHOD

- This method is applicable to those tasks which lack readily definable intermediate milestones and/or the effort/time required is very difficult to estimate.
- Example: Alignment of a major fan and motor may take from a few hours to a few days depending on the situation. Workers know when this work starts and when it is finished, but they never know the percentage completion in between.
- With this approach, one arbitrarily assigns a percent complete to the start of the task with 100% being recorded when it is finished.
- For short-duration tasks, one can assign a 50% completion with the start of the task
- With longer-duration activities something between 20-30% is typically used.

WAYS FOR MEASURING PROGRESS

SUPERVISOR OPINION

- This is a subjective approach and should be used for relatively minor tasks.
- Supervisor simply makes a judgment on the percent complete

II

WAYS FOR MEASURING PROGRESS

COST RATIO

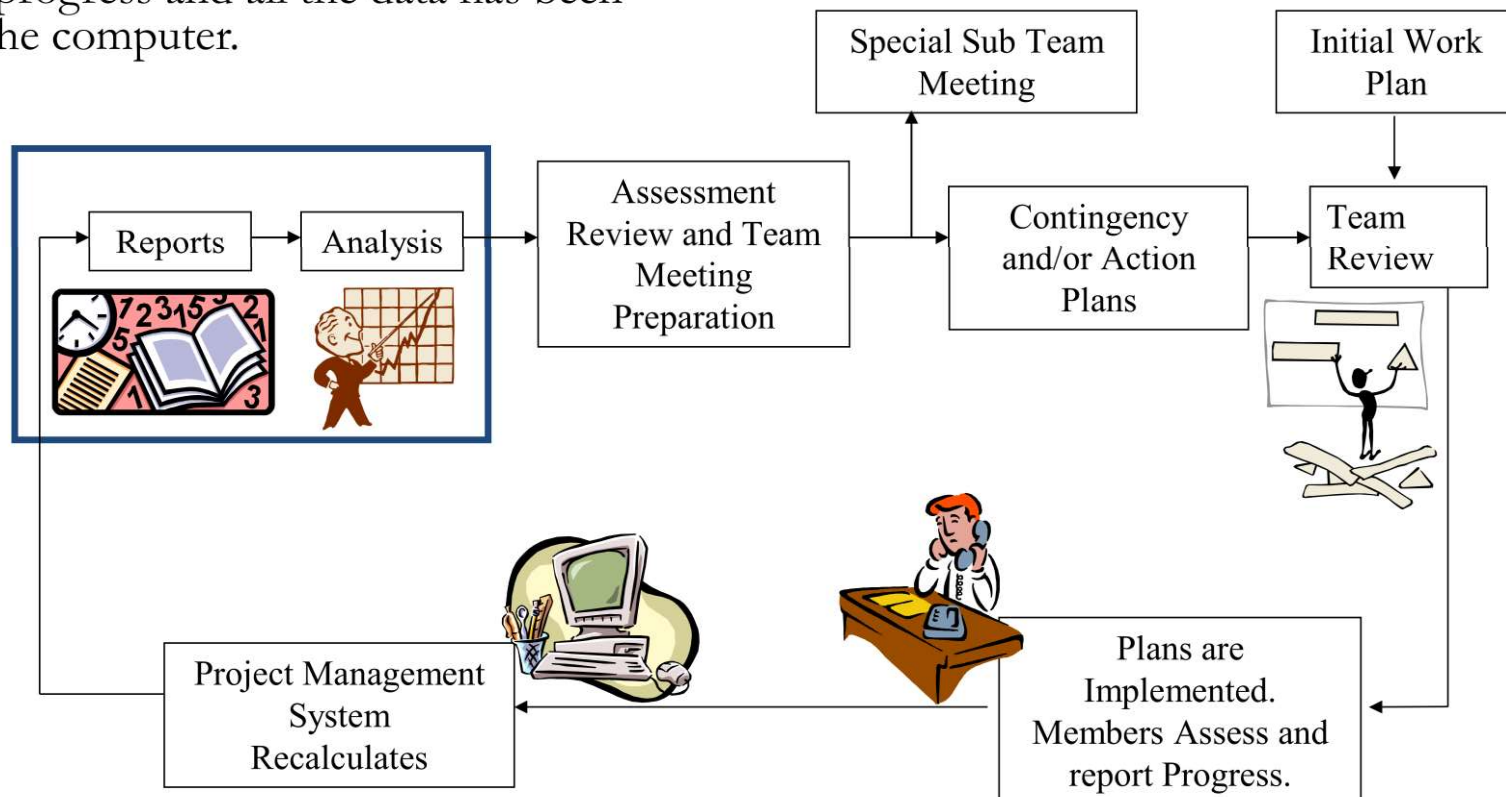
- Method is applicable to fields, such as project management, quality assurance, contract administration which involve a long period of time or are continuous during the life of a project.
- Such tasks are estimated and budgeted on bulk allocations of dollars or work-hours rather than on the basis of production.
- % Complete =
$$\frac{\text{Actual cost or Work-hours to date}}{\text{Forecast at completion}}$$

Generating Tracking Reports

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TRACKING REPORTS

- At this point, assume all task leaders have reported progress and all the data has been put into the computer.



TRACKING REPORTS

- All the progress data from individual task is fed into the computer.
- Looking at the entire project (sometimes 1000 to 2000 task) is not sufficient because it is so overwhelming. There is simply too much work to look at.
- We use tracking reports to focus in on areas of concern.

III

TRACKING REPORTS

- Different people will want to evaluate the project in different ways:
 - The team leader needs to look at the complete project. Area of concern may be applied labor and cost to date.
 - The task leader will be more concerned with a particular task and the immediate impact it has on other tasks. This person will be concerned with a subset of the total project.
 - An individual team member will want to evaluate the project from the perspective of activities for which they are responsible.
- Bringing the project back within schedule is accomplished with action and contingency plans.

III

TRACKING REPORTS

- Feedback Data Required to Control Project
- Many ways to examine project
 - Summarize complete project
 - Scrutinize complete project
 - Look at it from personal perspective
- Reports presents data required to:
 - Quickly understand what is expected
 - Quickly assess project / task status
 - Quickly surface issues
 - Justify requests for support
 - Highlight areas of concern
 - Formulate action and contingency plan

III

KEY REPORTS

- **Tracking Gantt** graphically show progress versus plan. Bar chart is the most flexible and commonly used tracking report.
- **Task reports** focus on specific tasks of interest
- **Applied labor reports** depict how much labor has been in relation to original estimates (actual labor used versus plan)
- **Cost reporting** is similar to labor reporting only we are tracking actual cost versus planned cost

III

KEY REPORTS

- **Logic Network (PERT Chart)** show task dependency and where the bottlenecks are in relation to what has been accomplished to date versus the plan
- **Resource Histogram** are used to evaluate how labor is being utilized
- **Open issues reports** allow to focus on areas of the project that are experiencing challenges
- **Project summary** reports only show the highest level of the project and are used for a large-scale view of the project

TASK REPORTS

- When analyzing the progress data, you may want to look at the project from various perspectives. We can sort and filter the project data in various ways to help us focus in on the information we need.
 - You can look at all tasks that are currently underway and those that are scheduled to start within the next two weeks.
- You may want to focus on just critical tasks.
- Task status
 - Not finished on schedule
 - Behind schedule
 - Ahead of schedule
 - Completed
 - Durations extend
- Filtering and/or Sorting
 - By project
 - By group
 - By individual

KEEPING PROJECTS ON TRACK

- Techniques for keeping projects on track
 - Focus on critical path
 - Look for inconsistencies
 - Look for opportunities to recover time

III

FOCUS ON CRITICAL PATH

III

- The project schedule will slip if an event on the critical path slips. Therefore, it is important to focus attention on it. Slippage or potential slippage of a critical path event should have the team's attention and receive high priority.
- Slippage impacts the project end date if task(s) is (are) critical:
 - Missing a task end date is serious
 - Create a mini-crisis, don't wait until the project end date is missed and be forced with a major crisis
 - Consider slippage of tasks on the critical path as reason to take drastic measures to correct the issue. Your project is late and deserves this type of attention. Consider working overtime, adding resources, or revise task dependencies to ensure that task will finish as scheduled
 - Look at tasks downstream to assess the impact a slippage will/might have on successor tasks and develop a plan to minimize the impact.
 - Have contingency plans ready ahead of time

FOCUS ON CRITICAL PATH

III

- You should constantly look for ways to improve performance on the critical path or “near critical path” (task(s) that will become critical if critical path is shortened slightly).
 - Any reduction in critical task duration results in a reduction in the overall project cycle or time that can be ranked to deal with unexpected events downstream.
 - Don’t squander opportunities to reduce cycle on the critical path, it will always pay dividends later on.
- Be proactive! Don’t wait until something happens to develop a plan to deal with an issue.
 - Invest some time at each team meeting thinking about how to deal with the high-risk events that might occur in the near term. Formal plans are not always required.
 - However, if a little time is invested up-front there is a lot less panic if disaster does strike.

LOOK FOR INCONSISTENCIES

- When looking over the data look for things that do not make sense:
 - If a great deal of effort has been applied to a task yet no significant progress has been made against the task. The team should probe to determine why more progress has not been made. People are often reluctant to ask for help, this is also a warning sign that the task event is likely to slip.
 - If very little effort has been applied and a great deal of progress has been reported. The team should probe to determine if there is an opportunity to shift resources to another event that is not on track.
 - If a task has been reported as complete always expect the deliverable to be available for a quality of event assessment. If it is not available, is the task truly complete? All team members must be aware of the fact that we are tracking task completion for a reason and that it does no good to report something complete when it is not. Better to tell me it is late than to tell me it is complete when it is not.

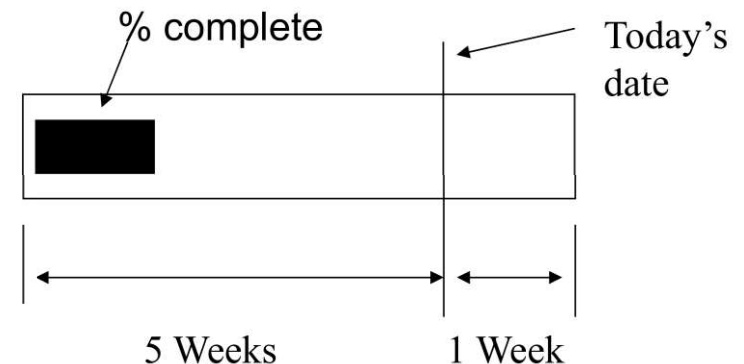
LOOK FOR INCONSISTENCIES

- Major changes in logic and/or effort signify that the original plan was in error or that the project scope has been changed.
 - First investigate the validity of these changes and ensure that the customer and/or management agrees that the changes are required.
 - Then look beyond the current area of involvement when adjusting the plans to accommodate these changes to ensure that other areas of the plan are not affected.

III

LOOK FOR INCONSISTENCIES

- Would you believe this task will be completed on time?
- If so, do you have any questions regarding quality of deliverables?

**Assume:**

A constant level of resource applied.

A linear effort required to complete task

LOOK FOR OPPORTUNITIES TO RECOVER TIME

- Turn to non-critical tasks to find resources to make-up the lost time
 - If **skills match** tasks in trouble
 - Delay start of non-critical tasks
 - Transfer resources to tasks that need support
 - If **skills do not match** task in trouble, we must be more creative
 - Build some slack downstream by looking for downstream critical path tasks that require skills that match those of non-critical tasks
 - Transfer resources to critical tasks downstream to recover time

III

LOOK FOR OPPORTUNITIES TO RECOVER TIME

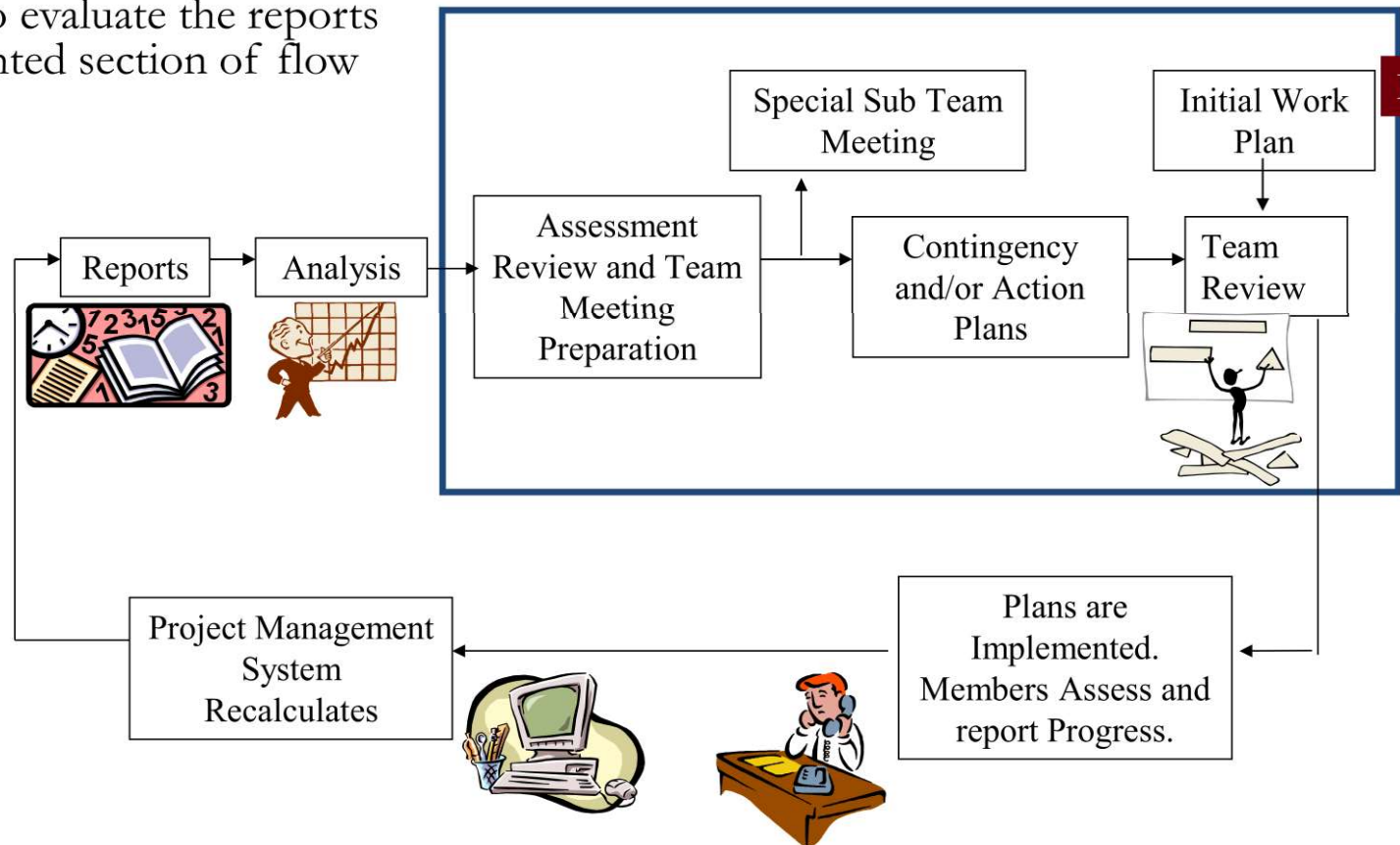
- When you anticipate a critical task will finish late, use a little overtime instead of adding resources before crisis hits:
 - Reinforces commitment to schedule
 - Reduces chance of burnout
 - Prevents the people burnout that often occurs due to excessive overtime towards the end of a project.
- Too often projects wait until the “big crisis hits” before they

Communications & Contingency Plans Development

Project Management – Project Monitoring & Control

COMMUNICATIONS & CONTINGENCY PLANS DEVELOPMENT

- The next step is to evaluate the reports (point out highlighted section of flow chart)



IV

INTERNAL COMMUNICATION OF PROJECT STATUS

IV

- Prior to a team meeting the project leadership:
 - Reviews and discusses project management data and reports
 - Determines critical issues to be recovered.
 - Focuses on key issues that will have a major impact on project
 - Not all issues need to be discussed in the team meeting
 - If an issue needs more in-depth analysis or action plan, a sub-team is formed
- Distributes an agenda for the team meeting.

INTERNAL COMMUNICATION OF PROJECT STATUS

- Sub-team meetings should be very focused and problem-solving oriented.
- A typical sub-team might include team members working on the tasks involved, subject matter experts and perhaps a meeting facilitator
- The major focus is to:
 - Define the issue,
 - Determine it's the root cause,
 - Assess the impact on the project and
 - Define a course of action.
- The output of the sub-team meeting is reported to the team during the team meeting for their review

IV

INTERNAL COMMUNICATION OF PROJECT STATUS

- Team meeting are focused on:
 - Communicating/reviewing status
 - Surfacing and resolving issues and gaining consensus on the course of action
 - Highly technical issues are typically not discussed in team meeting because of its cross-functional nature and number of participants

EFFECTIVE COMMUNICATION

- Communication is essential to the success of a project
- Project management is a communication enabler
 - Provides a focal point for communication
 - Everyone is “looking at the same play book”
 - Facilitates communication of progress, status and issues
- Project management does not ensure effective communication
- People are responsible for effective communication
- Effective meetings are essential

EFFECTIVE MEETINGS

- A meeting is an excellent tool for sharing information and building team consensus
- Preparation is critical to the success of a meeting
- Use the correct meeting format
 - Team Meeting
 - Working Session (Sub-team Meeting)
- Good meeting process should be observed
 - Adhere to effective communication precepts
 - Follow up with meeting minutes
 - Decisions reached
 - Actions items

ACTION PLAN

- Action plans are developed in response to unanticipated issues
- Hence, they are reactive rather than proactive plans
- Failure of test equipment during testing of a component is an example where an action plan might be developed

IV

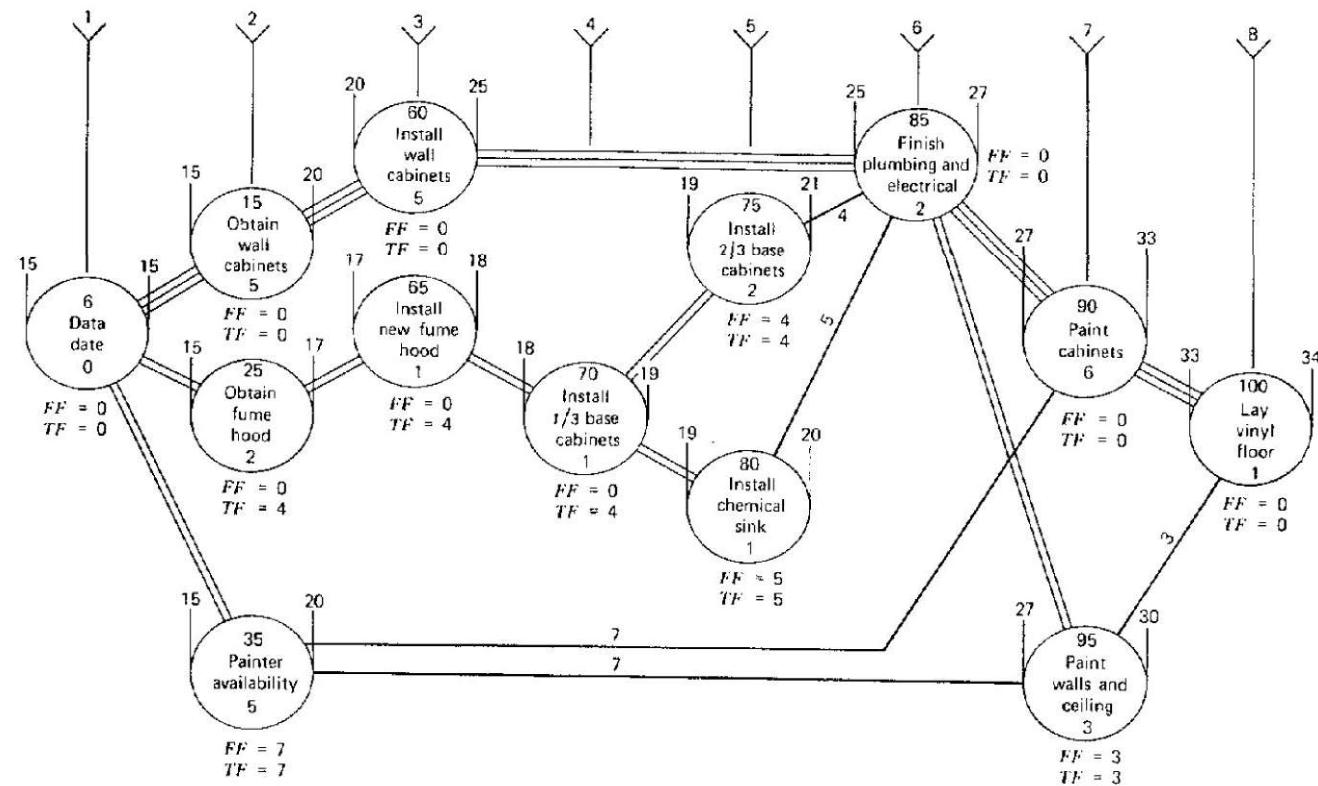
ACTION PLAN DEVELOPMENT

- Methodology
 - Analyze factors bearing on schedule such as resource availability, skills requirements, impact on project schedule (critical/non-critical task), cost impact of delay caused.
 - Identify and recommend alternatives (e.g., reschedule tasks, add resources, use of alternate methods)
 - Analyze impact of alternatives, additional cost, technical concerns, schedule risk
 - Choose “best” alternative
 - Secure management support and customer support if required
 - Most action plans will not require management approval and can simply be implemented by the team. Only plans that involve a great deal of risk and/or commitment of resources outside the team’s span of control require approval. However, we want to keep management informed of actions being taken.
 - Implement action plan
 - Modify work plan
 - Communicate changes to those affected
 - Monitor effectiveness

REMODELING CHEMICAL LAB EXAMPLE

UPDATED SCHEDULE

IV



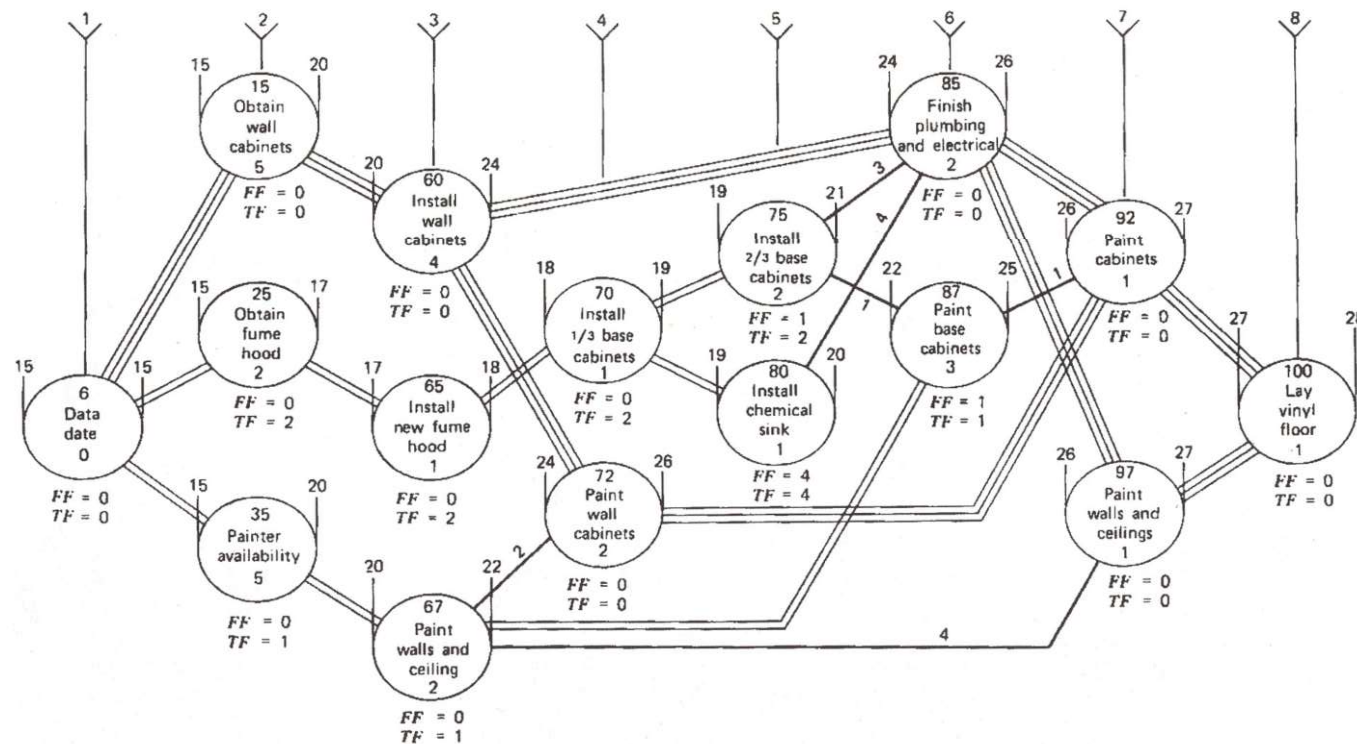
CHEMICAL LAB EXAMPLE CORRECTIVE ACTION (ACTION PLAN) PLAN

- Installation of the wall cabinets can be speeded up if we assign one additional carpenter to this task, and its duration will be reduced by one day.
 - This will shorten project duration by 1 day because this activity is critical
- Start painting walls and ceilings before cabinets are in place and before plumbers and electricians have completed their work
 - Break walls and ceilings into 2 activities and break paint cabinets into 3 activities: one for the wall cabinets, one for the base cabinets and one for touch up paint.
- Painters have one day to touch up around the cabinets and the plumbing and electrical fixtures after their installation

REMODELING CHEMICAL LAB EXAMPLE

REVISED SCHEDULE

IV



CONTINGENCY PLAN DEVELOPMENT

- When the risk of failure is high and the task or deliverable is critical to the success of the project, it is best to have a backup plan ready to go. The time and resource spent in the development and/pre-execution of a contingency plan is well spent.

IV

CONTINGENCY PLAN DEVELOPMENT

- Action plans are REACTIVE while Contingency plans are PROACTIVE
- Contingency plans are preplanned responses to high-risk events
 - Events critical to success of project
 - Events having high risk of failure
- Contingency plans
 - Cost \$ to develop
 - Develop only when necessary
- The methodology for developing contingency plans is very similar to that of action plans. However, contingency plans are developed proactively and are held in abeyance until needed.

CONTINGENCY PLAN DEVELOPMENT

- Example:

- A critical part to the design has never been manufactured before. There is risk involved if the part is not available by certain date. We should have an alternative to this part just in case we have problems making it and are not able to get it by the required date. We need a contingency plan to determine how we would deal with this risk if it should occur.

CONTINGENCY PLAN DEVELOPMENT

- Methodology is similar to developing action plans
 - Identify high risk events
 - Identify and recommend alternatives
 - To improve performance
 - To minimize schedule risks
 - To accommodate shift in requirements scope
 - Evaluate alternatives

CONTINGENCY PLAN DEVELOPMENT

- Key points:
 - Develop Contingency Plans when there is a high risk of failure on a critical task or event.
 - When developing a contingency plan, you follow a similar procedure to the one you use when creating a project plan define the mission, create a WBS...The major difference being that you have to look for ways to work the effort into the existing project plan.
 - Contingency plans are not typically added to the plan until the decision is made to execute them, You do not want to allocate resources to them and constrain your project with them until you decided they are necessary.
 - When developing contingency and/or action plans you should look at all the skills and resources that are available to the team.

Book Shelving Project Management Data

Project Management – Project Monitoring & Control

BOOK SHELVING PROJECT MANAGEMENT DATA

- Why do it?
 - Continuous improvement of process. Reduce cycle time and leverage past efforts
 - Learn from Past Experience. Data could be used on other projects to prepare improved estimates for resource requirements and costs.
 - Jump-start planning process

BOOK SHELVING PROJECT MANAGEMENT DATA

- What is in it?
 - Project management templates
 - Statistical data- effort and duration
 - Model work plans
 - Lessons learned
 - Alternative project plans (accepted and not accepted)
 - Benchmark data
- Statistical or quantitative information such as effort and duration data can be leveraged by future teams to help in planning and estimating

BOOK SHELVING PROJECT MANAGEMENT DATA

- How does it work?
 - Data submitted throughout the project
 - Major effort at the end of project
 - Not a garbage dump
 - Team effort
 - Submittals reviewed by “some committee”
 - Librarian catalogs and adds to bookshelf

Thank you for your attention!

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